

JOC Computational Project: Milestone 2 Report

Phase Objective: 2/5 — 3D Structural Parsing & Descriptor Extraction Matrices

Status: 100% COMPLETED

1. Physical-Organic Methodology & Justification

In Phase 2, the 3D Cartesian coordinates from each library file were systematically parsed. Two critical stereoelectronic dimensions were extracted mathematically: Maximum Molecular Dimension (D_max) in Angstroms and weighted substituent electronegativity trends using Pauling scaling tables. D_max isolates spatial steric parameters independent of subjective human estimation, while Electronegativity captures electronic background polarization. Together, these metrics supply the numerical independent variable framework requested by JOC regression guidelines.

2. Computed Descriptor Output Data Matrix

Compound ID	Substituent	D_max (Å)	Avg EN
1a	Methyl	3.54	2.35
1b	tert-Butyl	4.62	2.31
1c	Trifluoromethyl	4.394	3.113
1d	Phenyl	7.394	2.34
1e	p-Nitrophenyl	8.787	2.466
1f	p-Methoxyphenyl	9.72	2.383
1g	Cyano	4.962	2.578